Cheat Sheet Mid Terms

Signal, Systems and Transforms (Nanyang Technological University)



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Even Signal: $x(-t) = x(t)$ (e.g. cosine)
Odd Signal: $x(-t) = -x(t)$ (e.g. sine)
Periodic:

$$x[n+N] = x[n], \qquad N > 0$$

$$x(t+T) = x(t), \qquad T > 0$$

T is the period, T_0 : smallest period aka fundamental period.

$$f_0 = \frac{1}{T_0}, \quad \omega_0 = 2\pi f_0 = \frac{2\pi}{T_0}$$

$$x(t) = A\sin(\omega_0 t + \theta)$$

$$A: amp, \quad \omega_0: angular(radial) freq, T = \frac{2\pi}{|\omega_0|}, \theta: phase\ shift$$

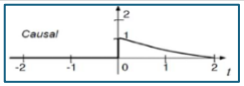
Which of the following signals are periodic ? Justify your ans

(a) $x_1(t) = \sin(10\pi t) + \cos(15\pi t)$
(b) $x_2(t) = \sin(10\pi t) + \cos(23t)$

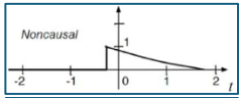
Answer (a):

- $\omega_1 = 10\pi = \frac{2\pi}{T_1} \rightarrow T_1 = \frac{1}{5}$ and $\omega_2 = 15\pi = \frac{2\pi}{T_2} \rightarrow T_2 = \frac{1}{7.5}$
- $\frac{T_1}{T_2} = \frac{1/5}{1/7.5} = \frac{3}{2}$ - a rational number and hence $x_1(t)$ is periodic.
- Fundamental period: $T_0 = 2T_1 = 3T_2 = 0.4$ sec
- Fundamental frequency: $f_0 = \frac{1}{T_0} = 2.5$ Hz

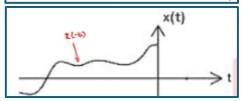
Causal Signal: signals are non-zero only for $t \geq 0$



Non-Causal Signals: signals are non-zero for some $t < 0$



Anti-Causal Signals: signals are non-zero only for $t \leq 0$



Energy Signal: has non-zero but finite energy for all time
Deterministic & non-periodic signals -> ENERGY signals.

$$0 < E_x < \infty, E_x = \int_{-\infty}^{\infty} |x(t)|^2\, dt$$

Power Signal: has finite but non-zero power for all time.
Have infinite energy. Periodic & random signals -> POWER signals.

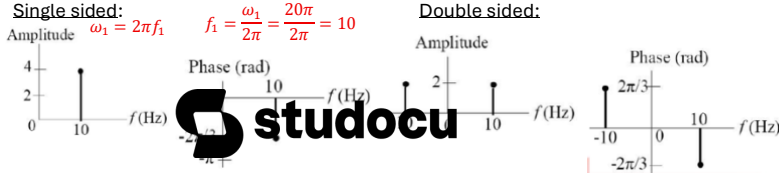
$$0 < P_x < \infty, P_x = \lim_{T\to\infty} \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} |x(t)|^2\, dt$$

Single-Sided Spect: $x(t) = \sum_{k=1}^{K} a_k \cos(\omega_k t + \theta_k)$, $\omega_k = 2\pi f_k$ **[only +ve freq]**
Double-Sided Spect: $x(t) = \sum_{k=1}^{K} a_k \cos(\omega_k t + \theta_k) = \sum_{k=1}^{K} \frac{a_k}{2}\left(e^{j(\omega_k t+\theta_k)} + e^{-j(\omega_k t+\theta_k)}\right)$
[both +ve & -ve freq]

Convert sine terms to cosine terms: $\boldsymbol{\sin\theta = \cos(\theta - \frac{\pi}{2})}$

$$e.g.\ x(t) = 2\sin\left(20\pi t - \frac{\pi}{6}\right) = 4\cos\left(20\pi t - \frac{2\pi}{3}\right), \quad A = 4, \quad phase: \frac{2\pi}{3}$$

Single sided: $\omega_1 = 2\pi f_1$ $f_1 = \frac{\omega_1}{2\pi} = \frac{20\pi}{2\pi} = 10$ Double sided:



Complex exponential signals: $x(t) = Ae^{at}$, $x[n] = Ae^{an}$
A and a are complex numbers.
Can be represented as a sum of sinusoidal functions.
General complex exponential system: $|A|e^{j\theta} = |A|e^{rt}e^{j(\omega t+\theta)} = |A|e^{rt}(\cos(\omega t+\theta) + j\sin(\omega t+\theta))$
Periodic exponentials/rotating phasors: $x(t) = |A|\cos(\omega t+\theta) + j|A|\sin(\omega t+\theta)) = Re[x(t)] + jIm[x(t)]$

$$|A|cos(\omega t+\theta) = |A|\frac{1}{2}\left(e^{j(\omega t+\theta)} + e^{-j(\omega t+\theta)}\right), \qquad |A|sin(\omega t+\theta) = |A|\frac{1}{2i}\left(e^{j(\omega t+\theta)} + e^{-j(\omega t+\theta)}\right)$$

Cross Correlation: find 2 different signals that are similar

$$R_{xy}(\tau) = x(t) \otimes y(t) = \int_{-\infty}^{+\infty} x(t)y(t-\tau)dt, \quad R_{xy}[k] = \sum_{n=-\infty}^{+\infty} x[n]y[n-k]$$

Autocorrelation: find when a signal is similar to itself delayed
It exhibits conjugate symmetry: $x(\tau) = x^*(-\tau)$

Autocorrelation function of an energy signal

- Consider an energy signal $x(t)$
- Autocorrelation of $x(t)$: $R_{xx}(\tau) = x(t)\otimes x(t) = \int_{-\infty}^{+\infty} x(t)x(t-\tau)dt$
- Autocorrelation function at the origin ($\tau = 0$) is equal to the energy of the signal (gives) $R_{xx}(0) == \int_{-\infty}^{+\infty} |x(t)|^2 dt = E_x$

Autocorrelation function of a periodic signal

- Consider a periodic signal $x(t)$ with period T_0
- Autocorrelation of $x(t)$: $R_{xx}(\tau) = x(t)\otimes x(t) = \frac{1}{T_0}\int_{-T_0/2}^{+T_0/2} x(t)x(t-\tau)dt$
- Autocorrelation function at the origin ($\tau = 0$) is equal to the power of the signal (gives) $R_{xx}(0) = \frac{1}{T_0}\int_{-T_0/2}^{+T_0/2} |x(t)|^2 dt$

Autocorrelation functions have the following properties:

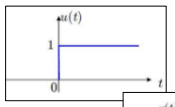
(i). Should be symmetrical $\left(R_x(\tau) = R_x(-\tau)\right)$

(ii). Maximum autocorrelation value occurs at the origin $\left(|R_x(\tau)| \leq R_x(0)\right)$

(iii). Auto correlation and power spectral density forms a Fourier transform pair $\left(R_x(\tau) \overset{FT}{\Leftrightarrow} S_x(f)\right)$ and power spectral density is always a non-negative function $(S_x(f) \geq 0)$

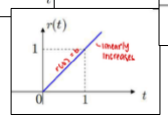
$g(t) period T_0$
$g(at) period \frac{T_0}{|a|}$

Unit Step Fn: $u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$



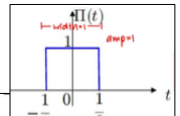
$T1-84: (X \geq 0)$
$T1-84: (X - a \geq 0), U(t-a)$

Unit-Ramp Fn: $r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$



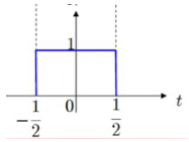
$T1-84: X * (X \geq 0)$ $r(t-a) = (t-a)u(t-a)$
$T1-84: (X-a) * (X \geq 0), r(t-a)$

Rect. Fn: $\Pi(t) = \begin{cases} 1, -\frac{1}{2} \leq t \leq \frac{1}{2} \\ 0, \ otherwise \end{cases}$



$T1-84: abs(X-3) \leq 0.5, \Pi(t-3)$

Sketching: $x(t) = x(\beta t + \alpha) = x\left[\beta\left(t + \frac{\alpha}{\beta}\right)\right] = x[\beta(t+t_0)]$



If $t_0 > 0 \rightarrow x(t)$ *is shifted left by* t_0
If $t_0 < 0 \rightarrow x(t)$ *is shifted right by* t_0
If $\beta < 0 \rightarrow x(t)$ *is reflected around the origin*
If $|\beta| > 1 \rightarrow x(t)$ *is compressed*
If $|\beta| < 1 \rightarrow x(t)$ *is expandedz* scaling factor: $\frac{1}{|\beta|}$

$\Pi(t) = u\left(t+\frac{1}{2}\right)u\left(\frac{1}{2}-t\right)$
$\Pi(t) = u\left(t+\frac{1}{2}\right) - u(t-\frac{1}{2})$

$\delta(t)$, *Dirac delta fn if it satisfies* $\int_{-\infty}^{+\infty} x(t)\delta(t)dt = x(0)$
Properties: $\int_{-\infty}^{+\infty}\delta(t)dt = 1$, $\delta(t) = 0\ for\ t \neq 0, \delta(at) = \frac{1}{|a|}\delta(t)$
Shifting Property:
$\int_{-\infty}^{+\infty} x(t)\delta(t-t_0)dt = x(t_0) \rightarrow \int_{-\infty}^{+\infty} x(t)\delta(\tau-t)dt = x(t)$
Kronecker delta fn: $\delta[n] = \begin{cases} 1, n = 0 \\ 0, n \neq 0 \end{cases}$ $\delta[an] = \delta[n]$ *(a is int)*
Properties: $\sum_{-\infty}^{+\infty} x[n]\delta[n]dt = x[0], \sum_{-\infty}^{+\infty} x[n]\delta[n-n_0]dt = x[n_0]$

Sampling: Time Domain Rep. (sampled signal-> discrete-time sig. s(t))

$$s(t) = x(t)\sum_{n=-\infty}^{\infty}\delta(t-nT_s) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t-nT_s) = x(t)\delta_{T_s}(t)$$

Sampling: Freq Domain Rep. (Fourier transforms->apply convolution)

$$S(f) = X(f)\otimes F[\delta_{T_s}(t)] \qquad f_s \geq 2W, if\ f_s < 2W: aliasing$$

$$= f_s\sum_{n=-\infty}^{\infty} X(f)\otimes\delta(f-nf_s) = f_s\sum_{n=-\infty}^{\infty} X(f-nf_s)$$

Natural Sampling: $s(t) = x(t)g_p(t)$
Flat-top sampling: $s(t) = \sum_{n=-\infty}^{\infty} x(nT_s)h(t-nT_s)$

Qns e.g.: Signal $x(t) = 10\cos\left(1200t + \frac{\pi}{6}\right) + 12\cos\left(3600t + \frac{\pi}{3}\right)$, uniformly sampled.

(a) Max sampling time interval:

$$\max signal\ freq(f_m), 2\pi f_m t = 3600t, f_m = \frac{3600}{2\pi} = 573Hz$$

$$sampling\ freq(f_s), f_s \geq 2f_m = 1146\frac{samples}{s}$$

$$T_s = \frac{1}{f_s} \leq 8.727 \times 10^{-4} s$$

(b) No. of samples for 1min of signal:

$$1146\frac{samples}{s} \times 60s = 68760\ samples$$

Addition and Subtraction Formulas

$$\sin(x+y) = \sin x\cos y + \cos x\sin y$$
$$\sin(x-y) = \sin x\cos y - \cos x\sin y$$
$$\cos(x+y) = \cos x\cos y - \sin x\sin y$$
$$\cos(x-y) = \cos x\cos y + \sin x\sin y$$
$$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x\tan y}$$
$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x\tan y}$$

Product-to-Sum Formulas

$$\sin x\cos y = \frac{1}{2}\left[\sin(x+y) + \sin(x-y)\right]$$
$$\cos x\sin y = \frac{1}{2}\left[\sin(x+y) - \sin(x-y)\right]$$
$$\cos x\cos y = \frac{1}{2}\left[\cos(x+y) + \cos(x-y)\right]$$
$$\sin x\sin y = -\frac{1}{2}\left[\cos(x+y) - \cos(x-y)\right]$$

$$\int \sin(ax+b)dx = -\frac{1}{a}\cos(ax+b) + C$$
$$\int \cos(ax+b)dx = \frac{1}{a}\sin(ax+b) + C$$
$$\int \sec^2(ax+b)dx = \frac{1}{a}\tan(ax+b) + C$$

Half-Angle Formulas

$$\sin^2 x = \frac{1-\cos 2x}{2}$$
$$\cos^2 x = \frac{1+\cos 2x}{2}$$
$$\tan^2 x = \frac{1-\cos 2x}{1+\cos 2x}$$

Trigonometric Identities

$$\sin^2 x + \cos^2 x = 1$$
$$1 + \tan^2 x = \frac{1}{\cos^2 x}$$
$$1 + \cot^2 x = \frac{1}{\sin^2 x}$$

Double-Angle Formulas

$$\sin 2x = 2\sin x\cos x$$
$$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1$$

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Output of a memoryless system at time t_0 depends on its input at the same time t_0: $U_{out}(t) = \frac{R_2}{R_1+R_2}U_{in}(t)$

Output of a system with memory depends on its past &/or future inputs: $y = 2x(t) + x^2(t-2)$

$y[n] = 1.05y[n-1] + x[n] \rightarrow y[n-1] = 1.05y[n-2] + x[n-1], y[n] = 1.1025y[n-2] + 1.05x[n-1] + x[n]$

Output of a causal system at time $t_0$depends only on its input at times $t < t_0$: $y[n] = 1.05y[n-1] + x[n]$

Output of a noncausal system at time $t_0$can depend on its input at future times $t > t_0$

A system is time-invariant if delaying the input by arbitrary $t_0$produces the output delayed by t_0

$y(t) = H[x(t)] \rightarrow \forall t_0, H[x(t-t_0)] = y(t-t_0)$

Power Rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad n \neq -1$$

$$\int x^{-1} dx = \ln|x| + C$$

Exponential

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \cos x\, dx = \sin x + C$$

$$\int \sin x\, dx = -\cos x + C$$

$$\int \sec^2 x\, dx = \tan x + C$$

$$\int \sec x \tan x\, dx = \sec x + C$$

$$\int \csc x \cot x\, dx = -\csc x + C$$

$$\int \csc^2 x\, dx = -\cot x + C$$

Linear Systems: adding/scaling inputs to the system results in adding/scaling the corresponding outputs.

For all constants a & input signals, $x(t)$, $H[ax(t)] = aH[x(t)]$ Homogeneity

For all input signals $x_1(t)$ & $x_2(t)$, $H[x_1(t) + x_2(t)] = H[x_1(t)] + H[x_2(t)]$ Additivity

Principle of Superposition: $H[a_1x_1(t) + a_2x_2(t)] = a_1H[x_1(t)] + a_2H[x_2(t)]$

Let $g(t)$ be an arbitrary function of time. Determine whether the system described by $y(t) = g(t)x(t)$ is linear and/or time-invariant.

Answer:

- Let $y_1(t)$ and $y_2(t)$ be the responses to $x_1(t)$ and $x_2(t)$.
- $y_1(t) = g(t)x_1(t)$ and $y_2(t) = g(t)x_2(t)$
- Multiplying these equations with a_1 and a_2 respectively and then add them up gives
- $a_1y_1(t) + a_2y_2(t) = a_1g(t)x_1(t) + a_2g(t)x_2(t) = g(t)(a_1x_1(t) + a_2x_2(t))$
- Let $y_0(t)$ be the response to $a_1x_1(t) + a_2x_2(t)$, then $y_0(t) = g(t)(a_1x_1(t) + a_2x_2(t))$
- Hence the system is **linear**
- The response to $x(t-t_0)$ is $g(t)x(t-t_0) \neq y(t-t_0) = g(t-t_0)x(t-t_0)$
- Hence the system is **time-variant**

Linear Time-Invariant Systems (LTI)

$$y(t) = \sum_{k=1}^{N} C_k x(t-t_k)$$

Any discrete-time signal can be represented as a linear combination of Kronecker delta functions.

$$x[n] = \sum_{-\infty}^{+\infty} x[k]\delta[n-k]$$

Apply $\delta[n]$ as input of the given system & obtain the output $h[n]$

$y[n] = H(x[n]), h[n] = H(\delta[n])$

Computing the output of a discrete-time LTI system:

Let $h[n]$ be the response of the system to $\delta[n]$, then $h[n] = H(\delta[n])$

Response to an arbitrary signal $x[n]$:

$$y[n] = H(x[n]) = H\left(\sum_{k=-\infty}^{+\infty} x[k]\delta[n-k]\right) = \sum_{k=-\infty}^{+\infty} H(x[k]\delta[n-k])$$

$$= \sum_{k=-\infty}^{+\infty} x[k]H(\delta[n-k]) = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$

Output of a LTI system when input is $x[n]$ is:

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{m=-\infty}^{+\infty} x[n-m]h[m]$$

Given the input signal $x[n]$, the output $y[n]$ of a discrete-time LTI system can be computed by convolving $x[n]$ with $h[n]$:

$$y[n] = h[n] * x[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{k=-\infty}^{+\infty} h[k]x[n-k]$$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} h[k]e^{-j\omega k} = |H(e^{j\omega})|e^{j\angle H(e^{j\omega})}$$

Any continuous-time signal can be represented as a finite weighted sum of shifted delta functions.

Computing the output of continuous-time LTI system:

Let $h[t]$ be the response of the system to $\delta(t)$, then $h(t) = H(\delta(t))$

Response to arbitrary signal $x(t)$:

$$y(t) = H(x(t)) = H\left(\int_{-\infty}^{+\infty} x(\tau)\,\delta(t-\tau)d\tau\right) = \int_{-\infty}^{+\infty} x(\tau)\,H(\delta(t-\tau))d\tau = \int_{-\infty}^{+\infty} x(\tau)\,h(t-\tau)d\tau$$

Given the input signal $x(t)$ the output of a continuous-time LTI system $y(t)$ can be computed by convolving $x(t)$ with $h(t)$:

$$y(t) = h(t) * x(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau$$

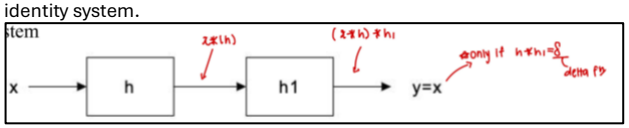
Properties of LTI systems:

1. Convolution n both continuous & discrete time is a commutative operation.

$$h[n] * x[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{k=-\infty}^{+\infty} h[k]x[n-k]$$

$$h(t) * x(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau$$

2. Distributive: $x * (h_1 + h_2) = x * h_1 + x * h_2$
3. Associative: $x * (h_1 * h_2) = (x * h_1) * h_2$
4. Invertibility: A system S is invertible iff there exists an inverse system S^{-1}s.t. $S^{-1}S$ is an identity system.



Here, the overall impulse response is $h_1$must satisfy for it to be the impulse response of the inverse system, namely $h * h_1 = \delta$

5. A system is stable if and only if the input produces a bounded output.

$y[n]$ is bounded in magnitude, hence is stable if $\sum_{-\infty}^{+\infty}|h[k]| < \infty$

For continuous-time LTI systems the stability condition is $\int_{-\infty}^{+\infty}|h(\tau)|d\tau < \infty$

Freq response of a LTI system:

Discrete-time LTI system

Consider a simple exponential signal $x[n] = e^{j\omega n}$, & $h[n]$ be the impulse response. The output of the LTI system:

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{k=-\infty}^{+\infty} h[k]x[n-k] = \sum_{k=-\infty}^{+\infty} h[k]e^{j\omega(n-k)}$$

$$= e^{j\omega(n-k)} \sum_{k=-\infty}^{+\infty} h[k]e^{j\omega(-k)} = H(e^{j\omega})e^{j\omega n}$$

Continuous-time LTI system

Consider an exponential signal $x(t) = e^{j[\omega t+\theta]}$

$$y(t) = \int_{-\infty}^{+\infty} x(t-\tau)h(\tau)d\tau$$

$$= \int_{-\infty}^{+\infty} e^{j(\omega(t-\tau)+\vartheta)}h(\tau)d\tau$$

$$= e^{j(\omega t+\theta)} \int_{-\infty}^{+\infty} e^{-j\omega\tau}h(\tau)d\tau = e^{j(\omega t+\theta)}H(\omega)$$

Response to an arbitrary signal

If the input signal is represented as $x(t) = \sum_{k=1}^{K} c_k e^{j(\omega_k t+\theta_k)}$

$$y(t) = H(x(t)) = H\left(\sum_{k=1}^{K} c_k e^{j(\omega_k t+\theta_k)}\right)$$

$$= \sum_{k=1}^{K} H(c_k e^{j(\omega_k t+\theta_k)})$$

$$= \sum_{k=1}^{K} c_k H(e^{j(\omega_k t+\theta_k)}) = \sum_{k=1}^{K} c_k H(\omega_K)e^{j(\omega_k t+\theta_k)}$$

$$= \sum_{k=1}^{K} C'_K e^{j(\omega_k t+\theta_k)}$$

Time-based approach

- Evaluate the impulse based response $h(t)$ by conducting an experiment or using system equation
- Find the output for an arbitrary input signal $x(t)$ by convolving it with the impulse response

$$y(t) = x(t) * h(t)$$

Frequency-based approach

- Represent the input signal $x(t)$ as a linear combination of sinusoidal signals

$$x(t) = \sum_{k=1}^{K} c_k e^{j(\omega_k t+\theta_k)}$$

- Compute the frequency response $\mathcal{H}(\omega)$
- The output is

$$y(t) = \sum_{k=1}^{K} C'_k e^{j(\omega$$

$\omega_k)$

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